

Partial Progress on Hamiltonian Policy-Improvement Paths in Multi-Action Discounted Markov Decision Processes

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Abstract

Goenka, Gupta, Khyalia, and Kalyanakrishnan ask whether the policy-improvement DAG of a discounted Markov decision process can contain a directed Hamiltonian path through all deterministic stationary policies when the number of actions is at least three. Equivalently, for an n -state, k -action MDP, can policy iteration be forced, by a suitable choice of improvement rule, to visit all k^n policies? This note gives partial progress only; it is not a solution of that problem and it does not construct the suggested 3-state, 3-action example visiting all 27 policies. The contributions are: an exact value-space reformulation in terms of isotone contraction hyperplanes, a linear feasibility certificate for checking any proposed path once transition kernels are fixed, and several rigorous obstructions to naive constructions. The obstructions rule out independent-state constructions, reset cycles inside a fixed one-coordinate fibre, and one-parameter affine coupling schemes that would need repeated reversals of action order.

Status. The results below should be read as a reduction and a set of certified special-purpose obstructions. They do not settle the Hamiltonian policy-improvement path problem for $k \geq 3$. In particular, they do not give a verified 3-state, 3-action MDP whose policy-improvement DAG contains a path through all 27 policies.

1 Introduction

For a finite discounted Markov decision process (MDP) with n states and k actions per state, there are k^n deterministic stationary policies. Policy iteration moves from a current policy to a locally improving policy, obtained by switching one or more states to actions with positive advantage. The possible one-step improvements define a directed acyclic graph, the policy-improvement DAG, whose vertices are policies.

For two actions, Melekopoglou and Condon constructed MDPs and policy-iteration variants that visit all 2^n policies. Goenka et al. ask whether an analogous “Hamiltonian” lower bound is possible when $k \geq 3$; a concrete first step would be a 3-state, 3-action MDP in which all 27 policies can be visited by policy iteration. Recent work of Agarwal, Wajid, and Kalyanakrishnan gives a larger lower bound for large action sets and proves the two-state case, but the full k^n question remains open in the literature.

The purpose of this note is narrower. It records what can be proved cleanly from a direct attack on the Hamiltonian question. The main point is that the problem has a precise geometric form: one must arrange nk Bellman hyperplanes, n parallel “families” of k hyperplanes each, subject to a strong monotone contraction constraint. A policy is the unique intersection obtained by choosing

one hyperplane from each family. A directed improvement edge is a strict sign condition at such an intersection. This reformulation is exact, and it explains both why the problem is finite-dimensional and why arbitrary Hamiltonian orientations of grids are not automatically realizable by MDPs.

2 MDPs, gains, and the PI-DAG

Fix $0 < \gamma < 1$. Let

$$\mathcal{S} = \{1, \dots, n\}, \quad \mathcal{A} = \{0, \dots, k-1\}.$$

A discounted MDP is specified by transition probabilities $T_{s,a,t} \geq 0$ with $\sum_t T_{s,a,t} = 1$, rewards $r_{s,a} \in \mathbb{R}$, and the discount factor γ . A deterministic stationary policy is a vector $\pi \in \mathcal{A}^{\mathcal{S}}$; its action at state s is denoted $\pi(s)$.

For a policy π , let P_π be the stochastic matrix whose s th row is $T_{s,\pi(s)}$, and let $r_\pi \in \mathbb{R}^n$ be the vector with $(r_\pi)_s = r_{s,\pi(s)}$. The value vector $V^\pi \in \mathbb{R}^n$ is the unique solution of

$$V^\pi = r_\pi + \gamma P_\pi V^\pi, \quad \text{equivalently} \quad V^\pi = (I - \gamma P_\pi)^{-1} r_\pi. \quad (1)$$

The inverse exists because the spectral radius of γP_π is at most $\gamma < 1$.

The action-value and gain of action a at state s under policy π are

$$Q^\pi(s, a) := r_{s,a} + \gamma \sum_{t=1}^n T_{s,a,t} V_t^\pi, \quad \rho^\pi(s, a) := Q^\pi(s, a) - V_s^\pi.$$

Thus $\rho^\pi(s, \pi(s)) = 0$.

Definition 1 (Policy-improvement edge). *For policies π, π' , let*

$$D(\pi, \pi') := \{s \in \mathcal{S} : \pi'(s) \neq \pi(s)\}.$$

There is a policy-improvement edge $\pi \rightarrow \pi'$ if $D(\pi, \pi')$ is nonempty and

$$\rho^\pi(s, \pi'(s)) > 0 \quad \text{for every } s \in D(\pi, \pi'). \quad (2)$$

The policy-improvement DAG, denoted $G(M)$, has vertex set $\mathcal{A}^{\mathcal{S}}$ and all edges satisfying (2).

The policy-improvement theorem implies that if $\pi \rightarrow \pi'$, then $V^{\pi'} \geq V^\pi$ coordinatewise and the inequality is strict in at least one coordinate; moreover $V_s^{\pi'} > V_s^\pi$ at every switched state. Hence $G(M)$ is acyclic.

Definition 2 (Hamiltonian PI-path property). *The pair (n, k) has the Hamiltonian PI-path property if there exist a discount factor and an n -state, k -action discounted MDP M such that $G(M)$ contains a directed path*

$$\pi^0 \rightarrow \pi^1 \rightarrow \dots \rightarrow \pi^{k^n-1}$$

listing all k^n policies exactly once.

3 Boundary cases and known results

The one-state case is trivial and is included only to fix the convention at the boundary.

Proposition 3 (One state). *For every $k \geq 1$, the pair $(1, k)$ has the Hamiltonian PI-path property.*

Proof. Let the single state be 1, let every action self-loop, and choose rewards $r_{1,0} < r_{1,1} < \dots < r_{1,k-1}$. If the current policy chooses action a , then $V^a = r_{1,a}/(1 - \gamma)$ and action b is improving if and only if $r_{1,b} > r_{1,a}$. Therefore

$$0 \rightarrow 1 \rightarrow \dots \rightarrow k - 1$$

is a directed Hamiltonian path. □

The two-state multi-action case is known and is not a contribution of this note.

Theorem 4 (Known two-state result). *For every $k \geq 2$, there is a two-state, k -action discounted MDP whose policy-improvement DAG contains a directed Hamiltonian path through all k^2 policies.*

This theorem is proved in the recent work of Agarwal, Wajid, and Kalyanakrishnan. It confirms the Hamiltonian property in dimension 2, but it does not give the 3-state, 3-action construction requested as the concrete next case of the Goenka–Gupta–Khyalia–Kalyanakrishnan problem.

4 Bellman hyperplanes and an exact geometric reformulation

For each state-action pair (s, a) define the affine Bellman hyperplane

$$H_{s,a} := \{v \in \mathbb{R}^n : v_s = r_{s,a} + \gamma \sum_{t=1}^n T_{s,a,t} v_t\}. \quad (3)$$

Also write

$$\Phi_{s,a}(v) := r_{s,a} + \gamma T_{s,a} v - v_s,$$

where $T_{s,a}$ is viewed as a row vector.

Lemma 5 (Policy values as hyperplane intersections). *For every policy π , the value vector V^π is the unique point in*

$$\bigcap_{s=1}^n H_{s,\pi(s)}.$$

Moreover, for policies π, π' , the edge $\pi \rightarrow \pi'$ is present in the PI-DAG if and only if $D(\pi, \pi') \neq \emptyset$ and

$$\Phi_{s,\pi'(s)}(V^\pi) > 0 \quad \text{for every } s \in D(\pi, \pi').$$

Proof. The equations defining the intersection are exactly the Bellman equations (1) for the policy π . Since $I - \gamma P_\pi$ is invertible, the intersection consists of the single point V^π . The second claim is just the definition of gain, because $\Phi_{s,a}(V^\pi) = \rho^\pi(s, a)$. □

Not every affine hyperplane of the form $v_s = \text{affine function of the other coordinates}$ can appear as an MDP Bellman hyperplane. The exact restriction is a nonnegative contraction condition.

Theorem 6 (Implementable Bellman graphs). *Fix $0 < \gamma < 1$ and a state s . An affine equation of the form*

$$v_s = c + \sum_{t \neq s} \beta_t v_t \quad (4)$$

can be represented as the Bellman hyperplane of some action at state s if and only if

$$\beta_t \geq 0 \quad (t \neq s), \quad \sum_{t \neq s} \beta_t \leq \gamma. \quad (5)$$

Proof. First suppose that (4) comes from a transition row $p = (p_1, \dots, p_n)$ and reward r , so

$$v_s = r + \gamma p_s v_s + \gamma \sum_{t \neq s} p_t v_t.$$

Since $1 - \gamma p_s > 0$, this rearranges to

$$v_s = \frac{r}{1 - \gamma p_s} + \sum_{t \neq s} \frac{\gamma p_t}{1 - \gamma p_s} v_t.$$

Thus $\beta_t = \gamma p_t / (1 - \gamma p_s) \geq 0$, and

$$\sum_{t \neq s} \beta_t = \frac{\gamma(1 - p_s)}{1 - \gamma p_s} \leq \gamma,$$

because $p_s(1 - \gamma) \geq 0$.

Conversely, assume (5). Put $B := \sum_{t \neq s} \beta_t$. Since $B \leq \gamma < 1$, define

$$p_s := \frac{\gamma - B}{\gamma(1 - B)}, \quad p_t := \frac{\beta_t(1 - \gamma p_s)}{\gamma} \quad (t \neq s), \quad r := c(1 - \gamma p_s).$$

The inequalities $0 \leq B \leq \gamma$ imply $0 \leq p_s \leq 1$ and $p_t \geq 0$. Moreover, the defining equation for p_s is equivalent to $B = \gamma(1 - p_s)/(1 - \gamma p_s)$, so

$$\sum_{t \neq s} p_t = \frac{B(1 - \gamma p_s)}{\gamma} = 1 - p_s.$$

Thus p is a stochastic row. Substituting these definitions into

$$v_s = r + \gamma p_s v_s + \gamma \sum_{t \neq s} p_t v_t$$

and dividing by $1 - \gamma p_s$ gives exactly (4). □

Corollary 7 (Exact geometric form of the problem). *Fix $0 < \gamma < 1$. The Hamiltonian PI-path problem is equivalent to the following hyperplane-arrangement problem. For every state-action pair (s, a) , choose an affine graph*

$$v_s = c_{s,a} + \sum_{t \neq s} \beta_{s,a,t} v_t, \quad \beta_{s,a,t} \geq 0, \quad \sum_{t \neq s} \beta_{s,a,t} \leq \gamma. \quad (6)$$

For each policy π , let V^π be the unique common solution of the n chosen graphs with actions $\pi(s)$. Then a Hamiltonian PI-path is exactly an ordering $\pi^0, \dots, \pi^{k^n-1}$ of all policies such that for every $m < k^n - 1$ and every changed state $s \in D(\pi^m, \pi^{m+1})$,

$$c_{s, \pi^{m+1}(s)} + \sum_{t \neq s} \beta_{s, \pi^{m+1}(s), t} V_t^{\pi^m} - V_s^{\pi^m} > 0.$$

Proof. By Theorem 6, the graphs satisfying (6) are exactly the Bellman hyperplanes of discounted MDP actions with discount γ . For a fixed policy, the map

$$F_\pi(v)_s = c_{s, \pi(s)} + \sum_{t \neq s} \beta_{s, \pi(s), t} v_t$$

has sup-norm Lipschitz constant at most γ , so it has a unique fixed point. Lemma 5 identifies this fixed point with the policy value and identifies the strict inequalities with improvement edges. $\square$

5 A linear certificate after transitions are fixed

The geometric formulation is still nonlinear if the transition probabilities are variables, because V^π depends on them. Once the transition kernels are fixed, however, the reward-realizability of a proposed path is an ordinary linear feasibility question.

Stack the rewards into a vector $r \in \mathbb{R}^{nk}$ indexed by pairs (s, a) . For a policy π , let $E_\pi \in \mathbb{R}^{n \times nk}$ be the selector matrix satisfying $E_\pi r = r_\pi$, and define

$$A_\pi := (I - \gamma P_\pi)^{-1} E_\pi. \quad (7)$$

Then $V^\pi = A_\pi r$. Let $e_{s,a} \in \mathbb{R}^{nk}$ denote the reward-coordinate unit vector for (s, a) , and let $e_s \in \mathbb{R}^n$ denote the state-coordinate unit vector.

Theorem 8 (Fixed-transition LP certificate). *Fix T and γ . Let*

$$\pi^0, \pi^1, \dots, \pi^{N-1}, \quad N = k^n,$$

be an ordering of all policies. There exist rewards making every step $\pi^m \rightarrow \pi^{m+1}$ a strict policy-improvement step if and only if the following finite strict homogeneous linear system is feasible in r :

$$\left(e_{s,a}^\top + \gamma T_{s,a} A_{\pi^m} - e_s^\top A_{\pi^m} \right) r > 0 \quad (8)$$

for every $m = 0, \dots, N-2$, every $s \in D(\pi^m, \pi^{m+1})$, and $a = \pi^{m+1}(s)$. Equivalently, strict feasibility is the same as feasibility of the closed linear system

$$\left(e_{s,a}^\top + \gamma T_{s,a} A_{\pi^m} - e_s^\top A_{\pi^m} \right) r \geq 1 \quad (9)$$

for all such triples (m, s, a) .

Proof. At policy π^m , the gain of switching state s to action $a = \pi^{m+1}(s)$ is

$$r_{s,a} + \gamma T_{s,a} V^{\pi^m} - V_s^{\pi^m}.$$

Substituting $V^{\pi^m} = A_{\pi^m} r$ gives the linear form displayed in (8). The proposed path is valid exactly when all these gains are positive.

It remains only to justify the margin formulation. There are finitely many strict inequalities. If r satisfies all of them, let δ be the minimum of the corresponding positive left-hand sides. Then $(1/\delta)r$ satisfies all inequalities with right-hand side at least 1. The converse is immediate. $\square$

Remark 9 (Verifiable infeasibility certificates). *Theorem 8 also explains what a rigorous computational nonexistence claim for a fixed transition template should provide. Let L be the matrix whose rows are the linear forms in (9). If $Lr \geq \mathbf{1}$ is infeasible, Farkas' lemma gives a certificate: a vector $y \geq 0$ with $y^\top L = 0$ and $y^\top \mathbf{1} > 0$. Such a certificate rules out the chosen transition template and proposed path, but not the original Hamiltonian problem, because a different transition template or a different path may still work.*

6 Rigorous obstructions to naive constructions

This section records several simple obstructions. None is strong enough to prove that Hamiltonian paths do not exist for $k \geq 3$; each only rules out a natural class of attempted constructions.

6.1 Independent states are far too weak

Proposition 10 (Independent self-loop states). *Suppose every action at state s self-loops to s , so $T_{s,a,s} = 1$ for all (s, a) . Then, after ordering the actions at each state by their rewards, every policy-improvement edge is coordinatewise nondecreasing in these fixed orders. Consequently every directed path in the PI-DAG visits at most*

$$1 + n(k - 1)$$

policies.

Proof. For a policy π , the value at state s is $V_s^\pi = r_{s,\pi(s)}/(1 - \gamma)$. Hence the gain of replacing $\pi(s)$ by a is

$$r_{s,a} + \gamma V_s^\pi - V_s^\pi = r_{s,a} - r_{s,\pi(s)}.$$

Thus action preferences at state s are fixed once and for all by the reward order at that state; they do not depend on the other coordinates of the policy. Along any improvement path, no coordinate can move downward in its fixed order. The rank

$$R(\pi) := \sum_{s=1}^n \text{rank}_s(\pi(s)),$$

where rank_s is the position of the action in the increasing reward order, strictly increases at every edge and lies between 0 and $n(k - 1)$. Therefore a directed path has at most $n(k - 1) + 1$ vertices. $\square$

6.2 No reset cycle inside a fixed fibre

A lexicographic base- k counter would need moves that look like $0 \rightarrow 1 \rightarrow \dots \rightarrow k - 1 \rightarrow 0$ in a low-order coordinate. Such a reset cannot be implemented while all other coordinates are fixed.

Proposition 11 (Fixed-fibre acyclicity). *Fix a state s and fix actions $\alpha(t)$ for all $t \neq s$. Consider the directed graph on the k policies*

$$\pi^a(s) = a, \quad \pi^a(t) = \alpha(t) \quad (t \neq s), \quad a \in \mathcal{A},$$

where an edge $a \rightarrow b$ is present when $\pi^a \rightarrow \pi^b$ is a policy-improvement edge. This graph is acyclic.

Proof. If $a \rightarrow b$, then the policy-improvement theorem applied to the one-state switch $\pi^a \rightarrow \pi^b$ gives

$$V_s^{\pi^b} > V_s^{\pi^a}.$$

Thus the real number $V_s^{\pi^a}$ is a strict potential function for edges in this fixed fibre. A directed cycle would strictly increase this potential and return to its starting value, which is impossible. $\square$

Corollary 12. *No Hamiltonian PI-path can be a literal lexicographic counter whose reset step changes only one state while all higher-order states are held fixed.*

6.3 One-dimensional affine context cannot reverse repeatedly

The preceding obstruction says that resets need context changes. A natural next attempt is to let the other states provide a context variable that changes the local order of actions. If this context is only one-dimensional and all local comparisons are affine in it, repeated reversals are still impossible.

Proposition 13 (One-parameter affine comparison obstruction). *Let $q_0, \dots, q_{k-1}$ be affine functions of one real variable, $q_a(z) = \alpha_a + \lambda_a z$. Fix two actions a, b . Along any monotone sequence of context values $z_0 \leq z_1 \leq \dots \leq z_m$, the sign of $q_a(z_i) - q_b(z_i)$ can change at most once, ignoring indices at which the sign is zero. Therefore no such one-parameter affine model can realize a prescribed sequence of strict action orders in which the same pair of actions switches relative order twice as the context moves monotonically.*

Proof. The difference

$$q_a(z) - q_b(z) = (\alpha_a - \alpha_b) + (\lambda_a - \lambda_b)z$$

is affine. If $\lambda_a = \lambda_b$, its sign is constant unless it is identically zero. If $\lambda_a \neq \lambda_b$, it has exactly one root. Hence on a monotone sequence of z values, the strict sign can pass from positive to negative or from negative to positive at most once. $\square$

This proposition does not rule out Hamiltonian PI-paths. It rules out only constructions in which all of the relevant context dependence at a state is compressed to a single monotone scalar and all pairwise comparisons are affine in that scalar. Corollary 7 suggests that any successful higher-dimensional construction must use genuinely multidimensional context or a more subtle nonmonotone traversal of context space.

6.4 The correct unique-sink orientation statement

It is tempting to replace the MDP problem by a purely combinatorial problem about orienting the product grid $K_k^{\square n}$. The following proposition states the safe version of that relationship and also records the necessary caveat about ties.

Proposition 14 (One-switch PI orientations and subgrids). *Restrict the PI-DAG to one-state switches. Let $\mathcal{B} = \prod_{s=1}^n B_s$ be any nonempty product subgrid, where $\emptyset \neq B_s \subseteq \mathcal{A}$. If the MDP restricted to the action sets B_s has a unique optimal policy, then the one-switch orientation on $\mathcal{B}$ has a unique sink. If this uniqueness condition holds for every subgrid, then the one-switch graph is an acyclic unique-sink orientation of the product grid. Without a no-ties or unique-optimum assumption, the unique-sink claim can fail.*

Proof. The one-switch graph is acyclic because every one-switch edge is a policy-improvement edge and policy improvement strictly improves the value vector in the partial order. A policy in $\mathcal{B}$ is a sink of the one-switch restriction exactly when no allowed one-state switch has positive gain. By the policy-improvement theorem applied to the restricted MDP with action sets B_s , this is equivalent to optimality in that restricted MDP. If the restricted optimum is unique, the sink is unique. Applying the same argument to every product subgrid gives the unique-sink property on every subgrid.

If the restricted MDP has two optimal policies, both have no positive allowed one-state improvement, so uniqueness of the sink may fail. Thus the no-ties hypothesis is not cosmetic; it is necessary for the literal unique-sink orientation statement. $\square$

The converse direction is false in general: an abstract acyclic unique-sink orientation of a product grid is not automatically an MDP construction. The Bellman hyperplanes must satisfy the isotone contraction constraints in Theorem 6. This is the main reason why existing Hamiltonian orientations of abstract grids do not by themselves settle the MDP problem.

7 What remains to be constructed

Corollary 7 leaves a concrete finite-dimensional task. For $n = 3$ and $k = 3$, one must choose nine affine graphs in $\mathbb{R}^3$,

$$v_s = c_{s,a} + \sum_{t \neq s} \beta_{s,a,t} v_t, \quad \beta_{s,a,t} \geq 0, \quad \sum_{t \neq s} \beta_{s,a,t} \leq \gamma,$$

and an ordering of the 27 policies such that the corresponding strict sign conditions hold at the 27 intersection points. A one-switch Hamiltonian path would require 26 strict inequalities; a path using multi-state switches would require one strict inequality for each switched state at each step.

The fixed-transition certificate in Theorem 8 gives a rigorous way to test a guessed transition structure and a guessed path. Passing such a test would produce an explicit MDP. Failing such a test only rules out that template, not the problem. The geometric formulation indicates why simple templates are unlikely to suffice: independent states have only linear-length paths, literal reset cycles are impossible within a fibre, and one-scalar affine couplings cannot repeatedly reverse the same pairwise action order.

8 Conclusion

This note obtains partial progress, not a solution. The proved results are an exact Bellman-hyperplane reformulation, an LP certificate for fixed transitions, and several obstructions to simple Hamiltonian constructions. These results do show that the uploaded attempted solution was aimed at a genuine problem in the authors' paper, but they do not establish the Hamiltonian PI-path property for $n \geq 3$, $k \geq 3$, nor do they give the suggested 3-state, 3-action example.

The most useful output of the analysis is the following sharply stated remaining problem: realize a Hamiltonian ordering of the product policy grid by intersection points of isotone contraction hyperplanes. Any future claimed construction can be checked either directly by the sign inequalities of Corollary 7 or, after transitions are fixed, by the linear certificate of Theorem 8.

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